

DP 024 (SET 2)
Physics 2
Semester II
Session 2024/2025
2 hours

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Fizik
Semester II
Sesi 2024/2025
2 jam



SES II
UNIT FIZIK
KOLEJ MATRIKULASI SELANGOR

PRA PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
PRE MATRICULATION PROGRAMME EXAMINATION

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DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

QUESTION	MARKS	SCORE
1	16	
2	12	
3	10	
4	14	
5	14	
6	14	
	80	

LIST OF SELECTED CONSTANT VALUES

Speed of light in vacuum	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space	μ_0	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space	ϵ_0	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude	e	$= 1.60 \times 10^{-19} \text{ C}$
Planck constant	h	$= 6.63 \times 10^{-34} \text{ J s}$
Electron mass	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$
Neutron mass	m_n	$= 1.674 \times 10^{-27} \text{ kg}$ $= 1.008665 \text{ u}$
Proton mass	m_p	$= 1.672 \times 10^{-27} \text{ kg}$ $= 1.007277 \text{ u}$
Hydrogen mass	m_H	$= 1.673 \times 10^{-27} \text{ kg}$ $= 1.007825 \text{ u}$
Deuteron mass	m_d	$= 3.34 \times 10^{-27} \text{ kg}$ $= 2.014102 \text{ u}$
Molar gas constant	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Avogadro constant	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant	k	$= 1.38 \times 10^{-23} \text{ J K}^{-1}$
Free-fall acceleration	g	$= 9.81 \text{ m s}^{-2}$
Atomic mass unit	1 u	$= 1.66 \times 10^{-27} \text{ kg}$ $= 931.5 \frac{\text{MeV}}{c^2}$
Electron volt	1 eV	$= 1.6 \times 10^{-19} \text{ J}$
Constant of proportionality for Coulomb's law	$k = \frac{1}{4\pi\epsilon_0}$	$= 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Atmospheric pressure	1 atm	$= 1.013 \times 10^5 \text{ Pa}$
Density of water	ρ_w	$= 1000 \text{ kg m}^{-3}$

LIST OF SELECTED FORMULAE

1.	$y = A \sin \omega t$	20.	$\tau = RC$
2.	$\omega = \frac{2\pi}{T} = 2\pi f$	21.	$Q = Q_o e^{\frac{-t}{RC}}$
3.	$v = A\omega \cos \omega t$	22.	$Q = Q_o \left(1 - e^{\frac{-t}{RC}}\right)$
4.	$a = -A\omega^2 \sin \omega t$	23.	$B = \frac{\mu_o I}{2\pi r}$
5.	$k = \frac{2\pi}{\lambda}$	24.	$B = \frac{\mu_o NI}{L}$
6.	$v = f\lambda$	25.	$B = \frac{\mu_o I}{2r}$
7.	$y(x, t) = A \sin(\omega t \pm kx)$	26.	$\phi = BA \cos \theta$
8.	$v = \frac{dy}{dt}$	27.	$\Phi = N \phi$
9.	$y = 2A \cos kx \sin \omega t$	28.	$\varepsilon = -\frac{d\phi}{dt}$
10.	$v = \sqrt{\frac{T}{\mu}}$	29.	$\varepsilon = Blv \sin \theta$
11.	$I = \frac{dQ}{dt}$	30.	$\varepsilon = -NA \frac{dB}{dt}$
12.	$Q = ne$	31.	$\varepsilon = -NB \frac{dA}{dt}$
13.	$\rho = \frac{RA}{l}$	32.	$R = 2f$
14.	$R_{eff} = R_1 + R_2 + R_3 + \dots + R_n$	33.	$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$
15.	$\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$	34.	$M = \frac{h_i}{h_o} = -\frac{v}{u}$
16.	$V = IR$	35.	$y_m = \frac{m\lambda D}{d}$
17.	$C = \frac{Q}{V}$	36.	$y_m = \frac{\left(m + \frac{1}{2}\right)\lambda D}{d}$
18.	$\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$	37.	$\Delta y = \frac{\lambda D}{d}$
19.	$C_{eff} = C_1 + C_2 + C_3 + \dots + C_n$		

Answer all questions.

1. (a) A progressive wave is represented by the following equation:

$$y = 0.03 \sin 2\pi(10t + x), \text{ where } x \text{ and } y \text{ are in meter and } t \text{ is in second.}$$

Determine the

- i. direction of the wave propagation.
- ii. amplitude of the wave.
- iii. wavelength.
- iv. frequency.
- v. wave propagation velocity.
- vi. particle velocity at $x = 1.0 \text{ m}$ and $t = 0.7 \text{ s}$.

[11 marks]

- (b) The equation of a stationary wave along a stretched string is given as

$$y = 0.44 \cos\left(\frac{1}{3}\pi x\right) \sin\left(\frac{1}{4}\pi t\right)$$

where y is in centimeter, x is in meter and t in second. Determine the

- i. wave number.
- ii. period of the wave.
- iii. amplitude of a single progressive wave.

[5 marks]

2. (a) A charge of 200 mC passes through a copper wire within 20 seconds , calculate the current that flows through the wire during that time.

[2 marks]

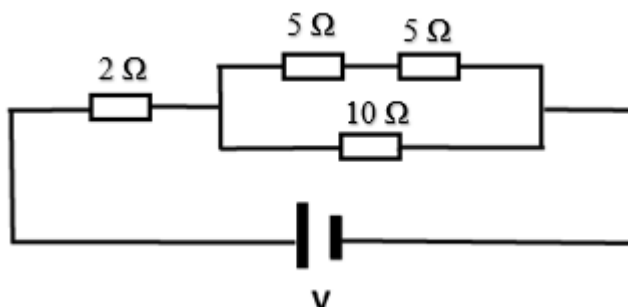
- (b) Calculate the resistance of an 5.0 m long aluminium wire with a diameter of 4.5 mm . Resistivity of aluminium at temperature 25°C is $2.72 \times 10^{-8} \Omega \text{ m}$.

[3 marks]

- (c) If a potential difference of 10 V is applied across a Nichrome wire with a resistance of 8Ω , what is the current flowing through the wire?

[2 marks]

- (d) Four resistors are arranged as shown in the figure below.

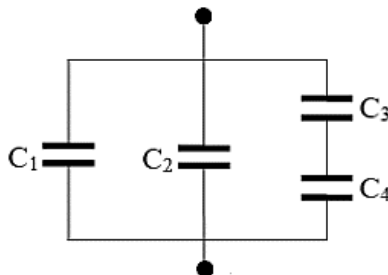


- i. Calculate the effective resistance of the circuit.
- ii. If the magnitude of V is 18 V, determine the amount of current flowing through the circuit.

[5 marks]

3. (a) Calculate the effective capacitance of the circuit in the following figure.

[Given $C_1 = 8 \mu\text{F}$, $C_2 = 21 \mu\text{F}$, $C_3 = 15 \mu\text{F}$ and $C_4 = 3 \mu\text{F}$]



[3 marks]

- (b) A fully charged capacitor has $6.0 \mu\text{F}$ and 3.50 mC charge. The capacitor has been discharged through a $30 \text{ k}\Omega$ resistor.

- i. Calculate the time constant for charging circuit.
- ii. Calculate time taken if just 30% charge left in the capacitor.
- iii. Sketch I-t graph to show discharging process.

[7 marks]

4. (a) A solenoid with 800 turns and a length of 1 meter carries a current of 4 mA. The solenoid is then cut into two equal parts, each with 400 turns. Calculate the magnetic field strength at the center of one of the solenoids after cutting.

[3 marks]

- (b) A metal coil with 50 turns and an area of 0.02 m^2 is placed in a magnetic field, which is perpendicular to the surface of the coil. The strength of the magnetic field changes from 8 T to 2 T in 2 seconds. If the resistance of the coil is 6Ω , determine

- i. the change in magnetic flux experienced by the coil.
- ii. the magnitude of the induced emf in the coil.
- iii. current that flows in the coil.

[7 marks]

- (c) A coil with 10 turns and an area of 0.5 m^2 is placed perpendicularly in a uniform magnetic field of 10 mT. The area of the loop is then reduced at a rate of $0.1 \text{ m}^2 \text{ s}^{-1}$. Calculate

- i. the magnitude of induced emf in the coil.
- ii. the resistance in the coil if the current flows through it is 1.2 mA.

[4 marks]

5. (a) An object with a height of 2 cm is placed 10 cm in front of a mirror. The mirror produces an image with a height of 4 cm. Determine the

- i. image distance.
- ii. radius of curvature of the mirror.
- iii. image distance if the object is placed 5 cm in front of the mirror.

[8 marks]

- (b) An object is placed in front of a converging lens with a focal length of 15 cm. An image is formed 60 cm behind the lens.

- i. Determine the object distance.
- ii. Sketch a ray diagram to show the formation of the image.
- iii. State 1 characteristic of the image formed.

[6 marks]

6. (a) In a Young's double slit experiment, an interference pattern is formed 1.00 m away from the double slits. The double slits are separated by 0.25 mm and the wavelength of light used is 550 nm. Determine the distance between the central bright to the

- i. 5th bright fringe.
- ii. 5th dark fringe.

[6 marks]

- (b) In a Young's double-slit experiment, light with a wavelength of 380 nm is used. A screen is positioned 200 cm away from the slits. The distance between the centre of the interference pattern and the first bright fringe is 0.15 cm.

- i. Calculate the separation of the slits.
- ii. What will happen to the size of fringes if the distance between the slits and the screen is decreased? Justify your answer.

[5 marks]

- (c) In a Young's double-slit experiment, the separation between the slits is 0.20 mm, and the screen is placed 2.0 m away from the slits. When monochromatic light is incident on the slits, the distance between the 9th and 10th bright fringes is 5.0 mm. Determine the wavelength of the light.

[3 marks]